

Polynomial Regression Report – BT2024066

Code repository: https://github.com/suhithreddy07/ML_assignment

1 Problems and data

var1 (turbine): predict the Net Power Score y from six settings $x_1, \dots, x_6$ (allowed degree ≤ 10). **var2 (thermal mapping):** predict the Thermal Anomaly Score y from three coordinates x_1, x_2, x_3 (degree ≤ 20). Each problem has 1000 training and 1000 test rows and no missing values. Metrics are MSE and R^2 .

All inputs lie in $[-1, 1]$ and are clipped at the edges (about 30% of var1 and 26% of var2 training values equal exactly ± 1). Linear correlation with y is weak (e.g. $|r| \leq 0.26$ for var1), so the signal is strongly nonlinear and includes interaction terms. Var1 has a distribution shift: test inputs lie farther from the origin than training inputs (mean distance 1.99 vs 1.74), so test predictions partly extrapolate toward the edges; var2 train and test distributions are more similar.

2 Method

For every degree d (1–7 for var1, 1–20 for var2), the full polynomial basis (all monomials of total degree $\leq d$, including cross terms) is built and standardised, and four estimators are compared: OLS, Ridge (L2), Lasso (L1) and ElasticNet (L1+L2). Penalties are tuned by 5-fold cross-validation on identical shuffled folds (seed 0). Polynomial expansion and scaling are re-fitted inside every fold, so there is no preprocessing leakage. The final model is refitted on all 1000 training rows.

3 Model comparison

Model	degree	var1		var2	degree	CV MSE	penalty
		CV MSE	penalty				
OLS	4	0.702	–	8	0.285	–	
Ridge	5	0.520	$\alpha = 17$	11	0.269	$\alpha = 1.19$	
Lasso	5	0.323	$\alpha = 0.0059$	11	0.278	$\alpha = 0.001$	
ElasticNet	5	0.334	$\alpha = 0.014, \ell_1 = 0.7$	11	0.270	$\alpha = 0.001, \ell_1 = 0.3$	

Table 1: Best 5-fold CV MSE of each estimator over the degrees tested.

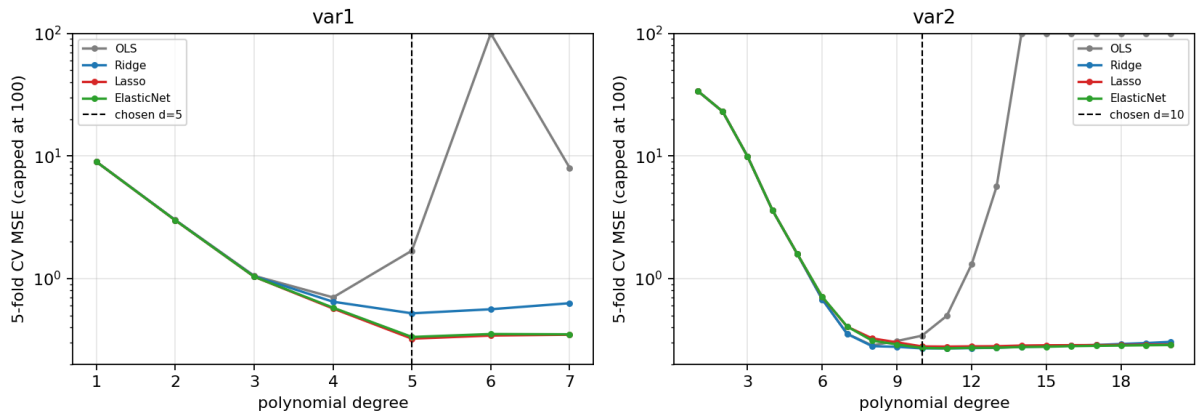


Figure 1: CV MSE of OLS, Ridge, Lasso and ElasticNet at every degree (log scale; OLS is capped at 100 for display). Dashed line marks the selected degree.

4 var1: Lasso, degree 5

Lasso CV MSE falls steeply with degree (8.96, 2.98, 1.04, 0.57, 0.32, 0.34, 0.35 for $d = 1, \dots, 7$) and is lowest at $d = 5$ before rising slightly. Low degrees underfit because the target depends on nonlinear interactions between

the input variables. Degrees above 7 were not fitted: the error was already flat-to-rising and degree 10 alone would create roughly 8000 polynomial terms for 1000 rows.

Lasso beats Ridge (0.32 vs 0.52) because the fitted signal is relatively sparse: only 116 of the 461 degree-5 terms have non-zero coefficients. ElasticNet gives no gain over Lasso, and OLS becomes unstable as degree increases (MSE 1.7 at $d = 5$ and about 154 at $d = 6$).

Shift check. Training on the 70% of rows nearest the origin and scoring on the outer 30% gives MSE 7.12, 2.17, 1.50, 0.53 and 1.13 for degrees 2–6, so degree 5 also performs best under this range-shift check. The final $\alpha = 0.01$ is a nearby value to the CV optimum (about 0.006); its CV error is essentially unchanged, and the degree-5 model is the best in the range-shift check.

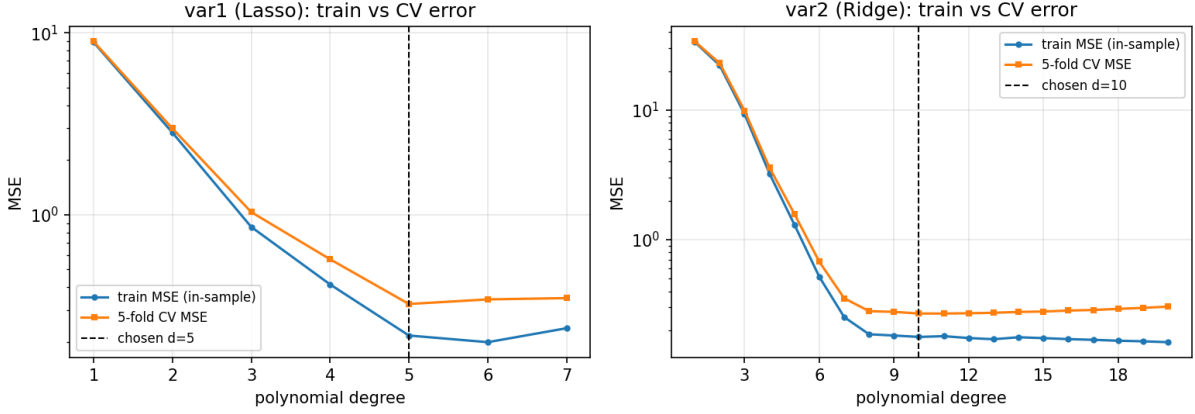


Figure 2: Train (in-sample) versus 5-fold CV MSE against degree for the chosen estimator type of each problem. The gap increases beyond the selected degree, indicating overfitting.

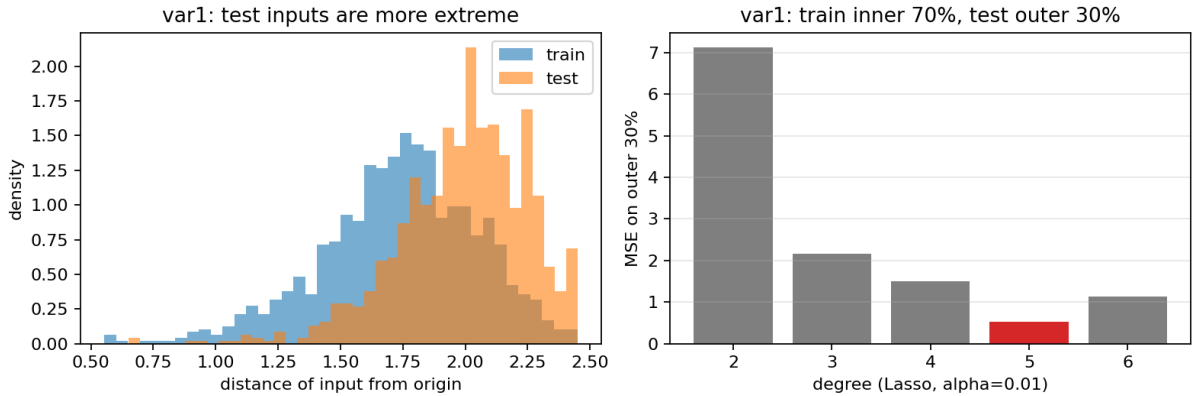


Figure 3: Var1. Left: test inputs are farther from the origin than training inputs. Right: MSE on the outer 30% of training data when trained on the inner 70%; degree 5 is best.

5 var2: Ridge, degree 10

CV MSE drops by about two orders of magnitude through the first several degrees and then flattens near 0.27, suggesting diminishing returns from increasing the degree. Ridge, ElasticNet and Lasso all reach this region between degrees 10 and 12, and Ridge is best or tied (0.269 in the single 5-fold comparison). With only three inputs, many polynomial terms contribute signal, so Ridge's smooth shrinkage is preferable to Lasso's term deletion.

OLS is stable only to about degree 8 and becomes numerically poor at higher degrees (MSE about 10^{11} at $d = 15$). Degree 11 is only 0.0002 better than degree 10, which is within the variation of the CV estimate, so the simpler degree 10 is retained. Predictions from degrees 8–12 differ from degree 10 by only about 0.06–0.10 RMS.

6 Diagnostics of the final models

Figures 4 and 5 use out-of-fold predictions. In both problems, actual-vs-predicted points follow the diagonal, residuals are centred near zero with no obvious curvature or funnel against the prediction, the residual histogram is approximately bell-shaped, the Q-Q plot is close to linear, and residuals show no clear pattern by row order. This indicates that the selected polynomial forms capture the main structure, with no obvious systematic residual pattern remaining.

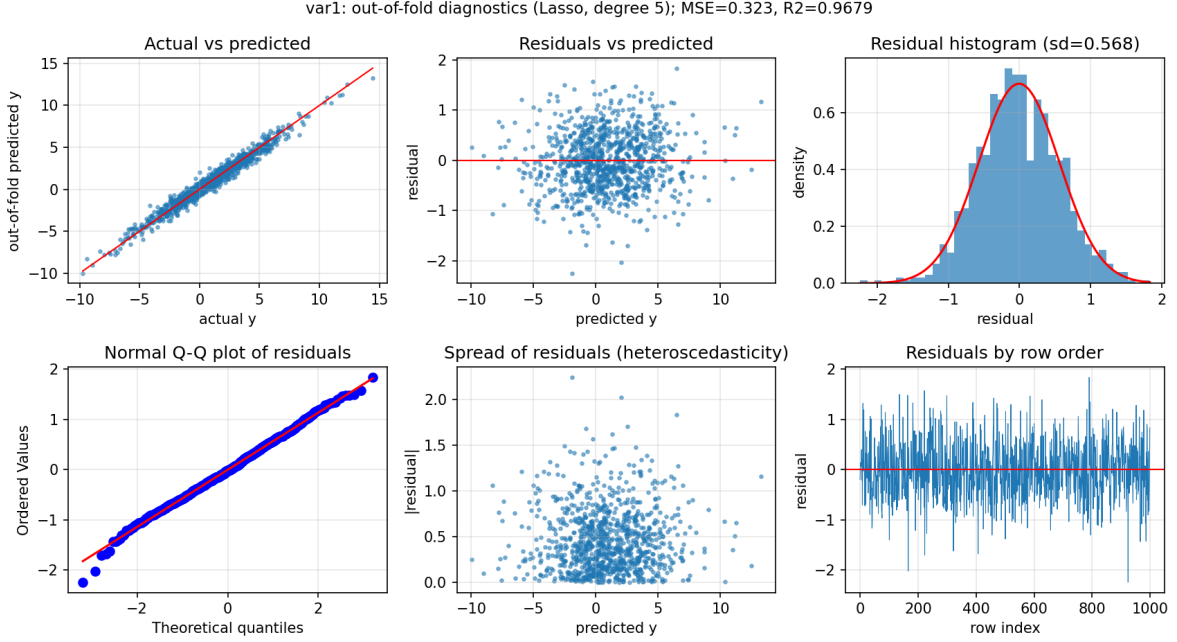


Figure 4: var1 (Lasso, $d = 5$): out-of-fold diagnostics.

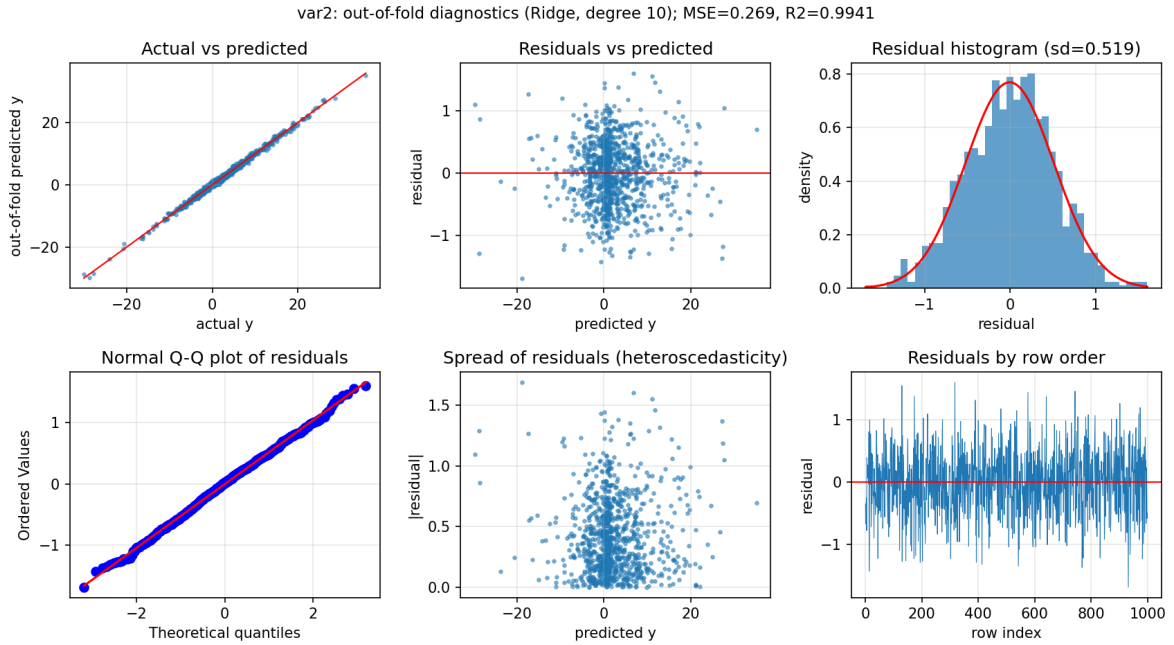


Figure 5: var2 (Ridge, $d = 10$): out-of-fold diagnostics.

7 Final models and expected performance

	var1 (turbine)	var2 (thermal map)
Final model	Lasso, $d = 5$, $\alpha = 0.01$	Ridge, $d = 10$, $\alpha = 1.0$
Polynomial terms	461 (116 non-zero)	285
Repeated-CV MSE (5×3)	0.326	0.272
Repeated-CV R^2	0.968	0.994
Training MSE	0.239	0.184

Expected test performance should be broadly close to the CV figures, although var1 may be somewhat worse because its test inputs lie nearer the edges of the training domain. Both prediction files contain one column y with 1000 rows in test-file order and no NaNs. The predictions remain broadly within the training-target scale, with some values extending slightly beyond the observed training range.

8 How to reproduce

Python 3.9+ is required. From inside the repository folder:

```
pip install -r requirements.txt
bash run_all.sh
```

Or run the steps individually:

```
python train_models.py
python compare_models.py
python shift_check.py
python make_figures.py
python diagnostics.py
```

The scripts produce the prediction CSVs, JSON/JSONL result files, and the final repository figures. To compile this report:

```
cd report
pdflatex report.tex
pdflatex report.tex
```